A MAP OF THE RANGE, NOT A WAY UP THE WALL: SIX WEEKS OF AI AGENTS ON 373 OPEN PROBLEMS IN ANALYSIS

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Abstract. For six weeks, one mathematician with a master's degree directed AI coding agents at every open problem that could be found in one area of analysis: Hardy, Bergman and Dirichlet-series spaces, Hankel and Helson operators, and Nehari-type theorems. The agents collected 373 problems, found that 64 of them had already been settled by others, and produced answers to 141 more. We classify these answers by how they were obtained and by how much they matter. Most are easy in hindsight: short arguments from known tools, results assembled from the literature, or counterexamples that are easy to check once found. Six are, in our judgement, of real significance. On the hardest problems, days of work by several agents produced maps of what fails, not proofs. We conclude that current agents are very good at mapping a field and clearing its long tail, and that a hard problem stays hard. All results of ours are sketches until they appear on arXiv or in a journal; this is a living document, updated as the people who asked the questions respond.

1. Introduction

Terence Tao has described research mathematics as a range of mountains:

> Tao gave the analogy of mathematicians trying to climb "a big mountain range with lots of tall mountains and lots of foothills". Humans can only climb one step at a time, but they can plan a route to the top of a mountain like Everest. Meanwhile, Tao said, current AIs are like jumping robots. They can sometimes "parkour their way to the top of a 6-foot wall" that a human couldn't climb. But they can't do long-term strategic planning. Those 6 feet might become 10 feet, or 100, Tao imagines, but "the little jumping robots are nowhere near the Mount Everests of math" [21].

This paper is a field report on what such robots do when pointed at an entire range. Over six weeks in autumn 2026 we used Claude Code and Codex agents to find, sort and attack every open problem we could locate in one research area. The agents found 373 problems in papers, problem lists and monographs. They established that 64 were already solved, and they produced answers to 141 others.

The answers need careful reading. Our main finding is that the gain was in cartography, not in climbing:

- The agents were very good at mapping a field: finding problems, classifying them, checking the literature, and clearing problems that were easy but had received little attention.
- Most of their answers are low-hanging fruit. Of the 141, we classify 6 as significant, 62 as useful, and 73 as minor.
- On problems where experts had made partial progress and stopped, the agents produced reformulations, special cases and catalogues of obstructions, not proofs.

These numbers need to be read with care. We did not solve 141 problems in the sense a mathematician would mean. The database mixes interesting questions from the field, questions at its research frontier, and questions we derived ourselves; many of the answers are to the last kind, or to questions that turned out to be easy once someone looked. Of almost 400 problems, about six of real significance were solved. What the agents added beyond that

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is partial: special cases and correct orders of growth for 142 problems, reductions of some 25 problems to three common hard cores, and written records of why each attempted route failed. These are new insights into the problems, not solutions of them.

Breadth explains the counts. Throw a hundred balls at a hundred baskets and some go in; that is useful, and different from getting better at the shot that matters. Tao's description of current tools as "artificial general cleverness" [24] fits what we saw.

We therefore present this method as the first step of a research project, not the last: a fast way to get to know a field and to find questions worth working on. Section 3 describes how the problems were found and sorted, Section 4 the workflow, Section 5 the results and their classification, Section 8 the hard problems, and Section 9 verification and credit. Section 10 places the work among other reports, and Section 11 lists its limits.

2. Setting

The area is the theory of Hardy and Bergman spaces on the disc and on the infinite polydisc, Dirichlet series viewed as functions on the infinite torus, and the Hankel, Helson and composition operators acting on them. A typical question asks for the sharp constant in an inequality, whether an extremal function can have zeros, or whether a bounded bilinear form has a bounded symbol. The active literature is a few hundred papers, much of it from a small number of research groups.

One person directed the work. He chose targets, judged whether a result mattered, steered agents that were stuck, read everything before it went out, and handled all contact with other mathematicians. The agents were Claude Code (Claude Opus 5.5 as the main session, with smaller Claude models for cheap subtasks) and Codex (several GPT models), often running at the same time in the same repository. The record of the work is the version history, with about 1000 commits, together with the logs: 14 Claude Code sessions with 393 subagents, and 586 Codex sessions, 420 of them subagents.

3. Building the map

The problems came from four sources, in this order.

(1) *Stated problems in a local paper collection.* Agents searched each paper for environments labelled Conjecture, Question, Problem or Open problem, and for phrases such as "remains open" and "it would be interesting to". Each candidate was recorded with the paper, the exact location, and a quote.
(2) *The papers those cite*, with the same extraction.
(3) *Synthesis passes.* Agents read the recent papers of the core authors "between the lines". This produced *implicit* questions, which a remark suggests without posing, and *inferences*: paper A proves one statement and paper B another, so a third should follow.
(4) *Problem lists and monographs*: workshop problem sessions, a classical problem book, and the open-problem sections of two monographs.

Each problem became one Markdown file with validated metadata. The metadata records:

- the theme, the kind (298 stated, 55 implicit, 20 inferences) and the origin (289 asked by others, 84 by us);
- tags, an impact score and a difficulty score;
- a dated literature check with quoted evidence;
- the stage reached in our own work;
- once anyone settled the problem, how far the resolution is trusted: journal, arXiv, or our own.

Moving from a single catalogue file to one file per problem let several agents and two different tools edit the database at once.

How difficulty was scored. Each problem received a difficulty score from 1 to 5, in two rounds. On 10 and 11 October, 194 problems from the first catalogue were assessed by cheap Codex agents (GPT-6 Luna), each with about ten minutes: a literature check, a short attempt, and a "speculative difficulty score and confidence" on the scale 1 = routine deduction or literature reconciliation, 2 = focused tractable project, 3 = substantial specialist project, 4 = probably requires a major new idea, 5 = longstanding frontier problem. Their brief warned that "a ten-minute failure to prove a statement is not evidence for a high score". On 12 October three Claude Opus 5.5s scored the whole database, each taking a group of themes, under a common brief that defined difficulty as "effort for us with current tools" and fixed an anchor problem at each level:

Difficulty	Meaning
1	short argument from known tools; days
2	focused effort; an existing method adapts, or a computation settles it
3	needs a genuinely new idea, but a route is visible
4	hard; the natural approaches are known to fail
5	major open problem, or equivalent to one

Each score came with a one-line justification. The agents read each record's statement, literature status and evidence, and the source where the statement was unclear; resolved problems were scored as originally posed. For the 189 problems that already had a score from the first round, the Opus agents kept it unchanged; they scored the rest from scratch. Over all 373 problems the initial scores were distributed as 37, 48, 110, 145 and 33 for difficulty 1 to 5. The scores moved during the work. After the scoring pass, 145 of the 168 problems that remained unsettled were re-rated, almost always upwards (127 by one level and 23 by two), as attempts failed; following the protocol, a problem that resisted its sessions was moved up. Of the 141 problems our agents settled, 38 were re-rated, mostly downwards. The final scores therefore record the outcome rather than predict it, and we use the initial scores of 12 October throughout. Even these are not blind: they were assigned after the directed work of 3 to 11 October, so for problems attempted then the score already reflects that attempt.

4. The workflow

Each problem was worked in sessions. A session begins by reading the record, the source passage and the log of earlier attempts, and by writing a three-line plan. It ends by rewriting a short *state of play* and appending one line to the *attempt log*: the route, the outcome, the reason, and the next step. A claimed result had to become a *solution sheet*: the problem as posed, the answer as a theorem, a complete proof with exact citations, and nothing else, built through a lint-gated LaTeX pipeline. A fresh agent with a fixed brief then refereed it. The referee re-read the question in its source, re-derived each step, reran any certificate, and decided whether the answer was full or partial.

On hard problems several agents worked in parallel *lanes*, each on one approach, with a coordinating session redirecting them. When lanes drifted into cataloguing obstructions, a *prove-mode* brief told each to commit to one strategy and push it to a theorem, or to a proof complete except for one precisely stated lemma. The full instructions are published with this paper as a reusable set of agent playbooks.

5. Results

5.1. **What the literature check found.** The difficulty assigned to a problem before anyone attempted it predicted the outcome well (Table 1). It also showed why easy problems are easy: nine of the twelve problems first rated easiest already had a published answer. Tao observed the same correlation on the Erdős problems: the problems current tools can solve are "precisely the category of nominally open problems that are most likely to have been

solved in the literature" [26]. The literature check also caught one problem recorded as resolved by a paper that had since been withdrawn.

Initial difficulty	Settled by others	Settled by us	Partial	Open
1	9	3	0	0
2	4	15	2	0
3	7	27	19	7
4	4	29	75	6
5	0	3	15	3

TABLE 1. Outcome by the difficulty assigned before any attempt, for the 275 records that had one.

5.2. **When the results came.** Figure 1 shows when the 141 answers were obtained, coloured by the initial difficulty. The database itself existed only from 12 October, so everything obtained in the directed work of 3 to 11 October appears in the first bar. That bar holds 30 answers, including four of the six we rate most significant. The bulk campaign of 12 October settled 45 problems, almost all of difficulty 1 to 3 and none of high significance. On 13 October the agents worked through harder problems: of its 66 answers, 25 were rated 4 or 5, and two are of high significance.

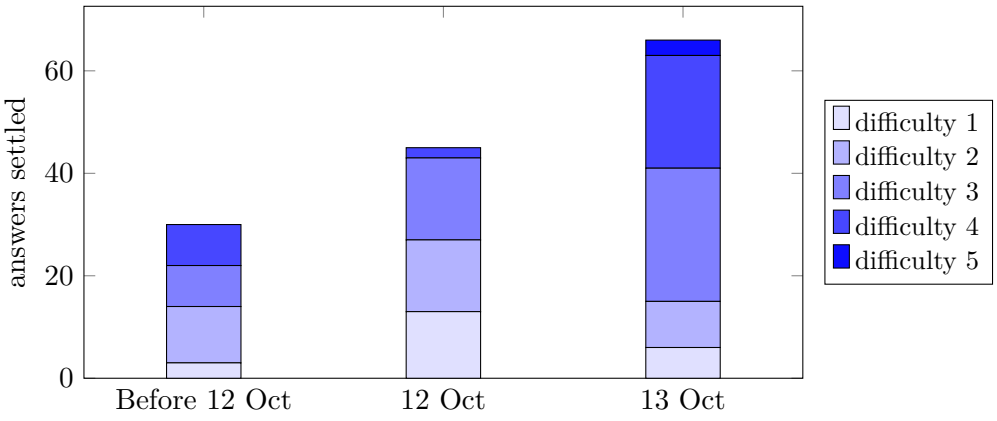


FIGURE 1. Answers settled by our agents, by when they were recorded and by the initial difficulty score. The first bar collects the directed work before the database existed; its dates come from the first appearance of each answer's write-up in the version history.

5.3. **What kind of results were these?** We classified each of the 141 answers along three axes.

- *Sign*: 82 proofs, 42 counterexamples, and 17 determinations of a value, set or order.
- *Kind*, meaning how the answer was obtained:
  - 20 assembled from the literature;
  - 58 routine new arguments;
  - 44 explicit constructions or certificates, which are the work to find and easy to check;
  - 19 substantial new arguments.
- *Significance*:
  - 6 high: paper-level answers to someone else's question;
  - 62 medium: full answers worth a short note;
  - 73 low: our own questions, conditional or elementary answers, and corollaries.

The classification is ours and preliminary; Table 2 gives the cross-tabulations.

	Literature	Routine	Construction	Substantial	Total
Proof	17	43	5	17	82
Counterexample	0	1	39	2	42
Determination	3	14	0	0	17

	High	Medium	Low	Total
Proof	4	33	45	82
Counterexample	2	24	16	42
Determination	0	5	12	17
Asked by others	6	57	26	89
Asked by us	0	5	47	52

TABLE 2. The 141 answers by sign and kind (top) and by sign, origin and significance (bottom).

Three things stand out.

- *Counterexamples are the natural strength.* Thirty-nine of the 42 counterexamples are explicit constructions or certificates. Searching a space of candidates and checking each one is exactly what agents do well, and what earlier work on machine-assisted discovery exploited [1, 13, 28].
- *Few proofs are substantial.* Only 17 of the 82 proofs needed a genuinely new argument; the rest are routine or assembled from the literature.
- *The significant results needed direction.* Of the six results we rate high, four are proofs and two are counterexamples. Three of the six came from focused sessions in which the human directed one problem with the asker's question in view, not from the bulk campaign over the database.

The six high-significance results are:

- that extremal functions for bounded linear functionals on H^p, $0 < p < 1$, are outer, answering a published conjecture in a stronger form, written jointly with the asker;
- a sharp bound on the number of zeros of extremals in a prescribed-coefficient problem;
- the finite-exponent limits of sharp Bernstein–Nikolskii constants;
- the exact order of growth of the H^1 norms of partial-sum operators for Dirichlet series;
- a counterexample to a positive-symbol Nehari question, with a computer certificate in outward-rounded arithmetic;
- that optimal polynomial approximants in H^p can vanish in the disc for every $p \neq 2$.

Each has been sent to the people who asked the question, or is being prepared for them. None is yet on arXiv.

6. Six discoveries, in the agents' words

The session logs record what each agent wrote when it reported a result. We quote them verbatim, with model and time (UTC); trims are marked [...] and our insertions are in square brackets. Table 3 gives the order.

Outer extremals. The first claim came after the director told the agent that its numerical sweeps did not get closer to a proof:

> I believe I've proved [the nonvanishing conjecture of Brevig and Saksman]: for $0 < p < 1$, every extremal for $C(k, p)$ has no zeros in the disc. [...] Only

Result	First reported	Found by	Sign
Outer extremals	8 Oct, 13:56	Claude Opus 5.5	proof
At most one zero	8 Oct, 20:49	GPT-6 Luna (Codex)	proof
Approximant zeros	10 Oct, 21:46	GPT-6.1 Sol; Claude	counterex.
Bernstein–Nikolskii limits	11 Oct, 19:13	Claude Opus 5.5	proof
H^1 partial sums	12 Oct, 22:21	GPT-6.1 Sol (Codex)	proof
Positive Nehari	12 Oct, 22:44	GPT-6 Astra	counterex.

TABLE 3. When the six high-significance results were first reported. Each was later checked by at least one other agent; the first two are also Lean-verified.

> I have checked the proof, plus numerics at $k = 2$; no one else has. —Claude Opus 5.5, 8 October, 13:56

Two hours later a subagent extended it from one coefficient to every bounded linear functional, including singular inner factors. The next evening a Codex agent estimated that a Lean formalization would take "months, potentially longer"; it was finished overnight:

> Done—the full theorem is now Lean-verified: every extremal for a nonzero bounded complex-linear functional on H^p, for every real $0 < p < 1$, is outer. —GPT-6.1 Sol, 10 October, 00:54

At most one zero. A Codex subagent found the key step; the deciding idea was to fix one perturbation phase rather than average over phases:

> The full real-direction Hessian yields a direct contradiction for any extremal with at least two zeros. [...] I checked the projections, the cancellation of the [...] terms, and admissibility of the perturbation. I found no gap in the argument. —GPT-6 Luna, 8 October, 20:49

The idea was relayed to the Claude session within minutes, which then reported that it had "reached the same conclusion independently". We cannot verify that claim, and we credit the Codex subagent.

Zeros of optimal polynomial approximants. This story took one evening. At 21:58 the director asked how close the agents were to the full result:

> We're not close to the full result if that means interior zeros for every $p \neq 2$. What we have is solid for p close to 1, and nothing yet for p near 2 or above it. —Claude Opus 5.5, 10 October, 21:58

Twenty-seven minutes later a subagent reported "there is no obstruction", and the main session wrote:

> Outer targets now work for $p > 2$ as well. Agent A found the construction, and I checked it independently. So for every $p \neq 2$ there is an outer target (even a polynomial one) whose degree-one OPA has a zero in $\mathbb{D}$. —Claude Opus 5.5, 10 October, 22:25

Bernstein–Nikolskii limits.

> [The finite-q problem] is proved: for every $s \geq 0$ and $0 < p < q < \infty$, $\lim A^{(s)}_{p,q,n} = D^{(s)}_{p,q}$, and the real version holds too. I checked the proof step by step; it has not been refereed. —Claude Opus 5.5, 11 October, 19:13

A second subagent then checked priority and found the problem still listed as open in a survey from two weeks earlier.

H^1 partial sums.

> The draft now proves a lower bound of order $\log N/\log\log N$ for every sufficiently large N, matching the published upper bound. The key step is turning product entropy into the missing L^1 output bound [...] subject to independent review. —GPT-6.1 Sol, 12 October, 22:21

A Claude referee later judged it "a genuine full-order answer to the $p = 1$ case of [an open problem of] Saksman–Seip" [22, Problem 4.5].

Positive Nehari. The same agent declined to count a numerical candidate before it had a certificate, and four minutes later had one:

> I'm not counting the new [...] numerical candidate as a breakthrough—it still needs a rigorous certificate. —GPT-6 Astra, 12 October, 22:40

> [The record] now has a rigorous counterexample certificate. For an explicit measure on two points in the real three-dimensional cube, the certified scalar norm exceeds the full Hilbert embedding norm. [...] This is a proved candidate awaiting independent review, rather than an 'almost proved' lead. —GPT-6 Astra, 12 October, 22:44

A Claude referee reran the certificate the next morning and found it "correct, no fixes".

The agents on their own work. The same logs show the agents assessing their limits frankly once challenged:

> I did a bounded first pass, not an all-out attempt. [...] I tested one route per open problem, often a relatively elementary construction, and usually stopped when it failed. [...] 14 failed routes do not demonstrate that I seriously exhausted the mathematical possibilities. —GPT-6 Astra, 13 October

> Against the original questions: 0 of these 4 fully solved. I made substantive partial progress, but selected easier subproblems instead of problems I could plausibly finish. —GPT-6 Astra, 13 October

> When everything you try collapses onto the same estimate, it usually means the problem needs a new idea, not more effort. —Claude Opus 5.5, 13 October

7. What the agents did not settle

Full answers are only half the picture. Of the 373 problems, 168 remain unsettled: 142 with partial results of ours and 26 without. We compared them with the 141 settled ones; the counts below come from the records, and the classification of obstructions is our judgement after reading each record's state of play.

7.1. How they differ.

	Unsettled (168)	Settled (141)
Initial difficulty 4 or 5	126	35
Asked by others	83%	63%
Impact 4 or more	19%	4%
Partial progress by others already in the literature	50%	21%
Answer is a counterexample (or expected to be one)	5%	30%
Inferences of ours	1	16
Mean number of working sessions	1.8	1.5

The problems that resisted were not neglected: they received slightly more working sessions than the settled ones, and only four received none. They differ in kind. They were asked by others, they matter more, half of them had already been attacked in the literature, and they ask for a characterisation, an exact constant or a uniform estimate rather than

a single construction. The settled problems, by contrast, include most of our own derived questions and most of the problems a counterexample could close. Themes rooted in number theory and probability (random multiplicative functions, Bohr radii, general Dirichlet series, Paley–Wiener extremals) were mostly unsettled; composition operators and the extremal problems around one published conjecture were mostly settled.

7.2. **What blocked them.**

Main obstruction	Records
Only special cases or families proved	59
Needs an estimate that is itself a known hard problem	37
Open-ended, ambiguous, or in need of reformulation	23
Endpoint or critical case; every method loses a log or a power	22
Exact constant wanted; only bounds or expansions found	14
Numerics suggest the answer, but no route to a proof	6
Depends on unrefereed or unavailable literature	5
Not attempted in depth	2

About 25 records reduce to three underlying problems: an L^1 endpoint estimate behind Schatten-class Nehari theorems, the value of one sharp constant in a multiplicative Hilbert-type inequality, and the unknown extremal functions of point evaluation in Paley–Wiener spaces. Settling any one of these would move many records at once; none moved.

7.3. **Full versus partial.** A full answer settles the question as posed; a partial one does not. Of the 142 partial results, the main type was:

Type of partial result	Records
Special case, subrange or family	90
Counterexample to a stronger statement or a literal reading	15
Correct order, or two-sided bounds, without the constant	13
Reduction to an equivalent reformulation	12
Conditional on a hypothesis or an unrefereed preprint	7
One direction only	5

A recurring kind is a local result near a point where the problem is already solved, for example near $p = 2$, which proves the conjecture in a neighbourhood but gives no uniform control away from it. This is the pattern the human director called "filling $\mathbb{R}$ with $\mathbb{Q}$": more and more special cases, none of which converges to the general statement.

8. Walls

We attacked several problems on which experts have made partial progress, for days and with several agents. Two examples, described without the technical details:

- *An endpoint Schatten-class Nehari problem.* Every route the agents found reduced to one square-function estimate in L^1 that is a theorem in its own right. They proved many special cases, wrote down equivalent reformulations, and catalogued exactly where each route failed. They did not find a proof, and the problem was handed back to the person who asked it.
- *A contractive restriction conjecture.* The agents proved the conjecture for two natural families of functions and refuted a stronger statement that would have implied it. The general case did not move.

In both cases the output was useful as a map. It is not progress on the wall itself, and we do not present it as such. The pattern matches Tao's image: some walls of six feet were climbed; the Everests were not approached.

9. Verification and credit

Producing a candidate proof was cheap; checking it was the bottleneck. This matches Tao's analysis of the problem-solving pipeline, in which generation is followed by verification, exposition, acceptance and absorption into the standard theory, and in which AI threatens "proof indigestion" [23]. Our answers were checked by agent referees, in some cases by a second model family, and in a few cases by Lean formalization or exact-arithmetic certificates. Agent referees caught real errors: a bound used outside its range, a missing hypothesis, and above all overstated scope, where an answer to a special case was reported as an answer to the question. They rarely asked whether the claim was the right one.

For this reason every result of ours is labelled a *sketch* until it appears on arXiv or in a journal, however it was checked. Before anything is posted, the people who asked the question are told; this follows the recommendations of the Leiden Declaration on attribution, disclosure and the humanity of authorship [18]. The questions belong to those who asked them. The proofs and computations were produced by AI agents and checked by the author. Following Tao's rule of thumb [23], no result should be published until a human can give a clear, expert-level talk on it.

10. Related work

Machine-assisted discovery. Machine learning has guided human intuition in knot theory and representation theory [7], and reinforcement learning has found counterexamples to conjectures in combinatorics [28]. Language models coupled with evaluators have found new constructions [1, 13], and have been applied to many problems at once [14]. Our counterexamples are of this kind: constructions that are hard to find and easy to check.

Formal proof. Language models for theorem proving date back at least to [17, 20]; olympiad geometry has been solved without human demonstrations [2]. The mathlib library [19] underlies much of this, and Avigad discusses the wider "formal turn" [3].

Evaluation. Early evaluations of language models on mathematics include [6, 11]. Benchmarks with unpublished problems [12, 16] and the First Proof project, which has experts referee AI solutions to unpublished research problems [10], measure capability under controlled conditions. Our report is neither controlled nor a benchmark; it is one practitioner's account of a whole area.

Research practice. Experiments with frontier models as research assistants are collected in [5]. The Erdős problems database [8] and its AI wiki [9] are the closest analogue to our problem database; the wiki warns that "absence of past progress may reflect obscurity rather than difficulty". On the goals of mathematics beyond problem solving we follow Thurston [27], Gowers [15], Tao [23, 25] and Bessis [4]. Journalism on the current state of the field includes [21], where Igor Pak notes that language models are "fantastic in finding the right references".

11. Limits

This is one person's experience in one area, over six weeks, without a control group. The difficulty and significance scores were assigned by the same agents and by the author, not by independent experts. Correctness has not been audited on a random sample; we intend to do so, and the counts above should be read as counts of claimed answers. The models will be superseded within months. Finally, the area has a dense literature of recent preprints, which made the literature check unusually productive.

Tool and computational resource disclosure

The work used Claude Code from Anthropic, with Claude Opus 5.5, Sonnet and Haiku, and Codex from OpenAI, with several GPT models; Python with mpmath, SymPy and cvxpy for numerics and certificates, Lean for one formalization, and LaTeX built through a lint gate.

This paper was drafted with Claude Code from the project's records and logs and revised by the author. The agent instructions, templates and tools are available in the accompanying process repository.

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